

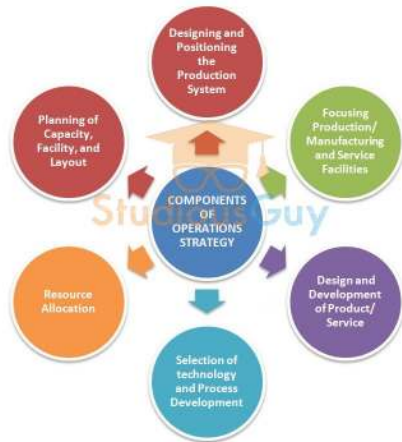
Business Study Notes

Topic 1 - Operations

Outcomes:

- H1 critically analyses the role of business in Australia and globally
- H2 evaluates management strategies in response to changes in internal and external influences.
- H3 discusses the social and ethical responsibilities of management
- H4 analyses business functions and processes in large and global businesses
- H5 explains management strategies and their impact on businesses
- H6 evaluates the effectiveness of management in the performance of businesses
- H7 plans and conducts investigations into contemporary business issues
- H8 organises and evaluates information for actual and hypothetical business situations
- H9 communicates business information, issues and concepts in appropriate formats

Learn About:	Syllabus Points:	Summary Notes:
Role of Operations Management	Strategic Role of Operations	Operations: The business processes that involve the conversion of inputs (transformed and transforming) into final goods and services, known as outputs. Strategic: A long-term direction for all key business operations. During the operations process, businesses convert inputs into outputs and are also involved in creating value. Value adding refers to the added value as inputs are transformed into outputs. <u>Operations managers focus on achieving:</u> - Cost Leadership - Product differentiation Cost Leadership: Businesses aiming for the lowest possible costs or the most price-competitive



market. Businesses need to aim to streamline costs across the whole operations process. Allows higher profits and a competitive advantage

Reducing cost methods:

- Exploit economies of scale
- Produce standardised goods
- Engage in supplier rationalisation
- Outsource materials and labour

Product differentiation: Involves businesses distinguishing their products from their competitors. They do this by changing product features, varying product quality or including additional benefits. For services, they improve the level of expertise brought to a service, the time or the qualifications of a service provider.

Standardised goods: Mass-produced goods are uniform in quality and meet predetermined levels of quality.

Customised goods: Goods tailored to the needs and wants of the customer.

Operations	
Finance	Ensures there are adequate funds to carry out transformation processes. Financial transactions reflect the true value of the business.
Marketing	Ensures the business meets the needs and wants of customers through market research.
Human Resources	Effectively deals with employees and employment issues. Deals with employment resourcing, engaging in activities of

		acquisition, development, maintenance and separation.															
	Goods and Services in Different Industries  Goods vs. Services 	Products may be classified as a good or a service: <table border="1"> <thead> <tr> <th>Characteristics</th><th>Good</th><th>Service</th></tr> </thead> <tbody> <tr> <td>Tangibility</td><td>All goods are tangible (seen, moved, touched)</td><td>Services are intangible - they only exist when they are being performed.</td></tr> <tr> <td>Customisation</td><td>Tend to be standardised</td><td>Are customised (haircut), may be standardised to a high degree (escape rooms)</td></tr> <tr> <td>Ownership</td><td>Can be owned and transferred between people</td><td>It cannot be owned</td></tr> <tr> <td>Value</td><td>Easier because you can find out how much each part costs (labour, materials).</td><td>More subjective (level of skill, experience, ability)</td></tr> </tbody> </table> For both goods and services, differentiation can come from cross-branding or strategic alliance (Coles and Shell): which adds value by offering consumers added benefits. The operations function within any business will be shaped by the range and types of goods/services that are produced. For goods: Standardised: Mass-produced on an assembly line, used in everyday retail products (clothes,	Characteristics	Good	Service	Tangibility	All goods are tangible (seen, moved, touched)	Services are intangible - they only exist when they are being performed.	Customisation	Tend to be standardised	Are customised (haircut), may be standardised to a high degree (escape rooms)	Ownership	Can be owned and transferred between people	It cannot be owned	Value	Easier because you can find out how much each part costs (labour, materials).	More subjective (level of skill, experience, ability)
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		shoes, furniture) 2. Customised: Varied according to customer needs, usually luxury products (luxury bags, customised phone cases) OR 1. Perishable: Short shelf-life, e.g. veggies and meat. 2. Non-Perishable: Last a long time, e.g, tools, cars. For services: 1. Standardised: fast food, cheap/quick services, e.g. Maccas. 2. Customised: Most are customised, doctors, lawyers, and professional nails. Intermediate goods have gone through one set of operational processes and then become inputs into further processing (iron ore - screws - table)
	Interdependence 	Interdependence: mutual dependence that the key business functions have on one another. The key business functions (marketing, operations, etc.) are interdependent — each relies on the other to perform effectively. Operations ↔ Marketing Marketing determines the type of design through market research of a good, which impacts operations through sourcing the materials and creating the process to produce the good. Operations ↔ Finance Finance is the function that records and summarises financial information. Operations make up a large portion of costs for a business. We need to know these costs to make decisions (reduce costs to increase profits). Operations ↔ Human Resources People are recognised as the most important resource to a business. The nature of work in the

operations function (builders, managers) determines the types of skills and people the business needs.

CASE STUDY - Virgin Australia

The strategic role of operations management involves operations managers contributing to the strategic direction or plan of the business which as seen in Virgin Australia.

- Virgin decided to acquire other businesses (such as Tiger Australia) in 2013 which directly impacted other key business functions (finance, hr, marketing). Saved \$200 million in operational efficiencies.
- Then restructured/simplified their operations due to COVID-19.
- Provided sufficient financial resources and gradually returned to the aviation industry.

CASE STUDY - Samsung

Samsung used a cost leadership strategy by reducing its labour, capital, taxation, and environmental costs by moving its production from Korea and China to Vietnam in 2016.

- Allowed a reduction in the final price of their phone.

CASE STUDY - Aldi

Aldi carefully planned their operations as the manager knew strategic growth would take about 10 years. Long-term growth meant re-investment into supply lines and building warehouses.

- Attracted loyal customers through its 'special buys' promotions.
- Operations have to work closely with marketing and finance in sourcing what customers desire at a price they deem attractive.
- Interdependence of key business functions continues to build Aldi's brand in Australia.

Influences

Globalisation, Technology,
Quality Expectations,
cost-based competition,
government policies



Globalisation: The removal of barriers to trade between nations and increasing integration of countries (transfer of capital, ideas, and customers).

- Has a significant influence on the operations function as businesses are orienting their practices towards the global market (targeting global consumers, production, and technology).
- Globalisation often influences production regarding location (cheaper country)
- Maccas operates in 119 countries and as a result, has to cater for customer's tastes in different countries.

Technology: The design, construction, and/or application of devices, methods, and machinery to operational processes (i.e. the use of robots in manufacturing).

Examples include:

1. Office technologies (laptops, scanners)
2. Software (word processing, spreadsheets)
3. Machinery used in manufacturing (robotics, complex machines)
4. CAS (Computer-aided design) to create 3D designs

Quality expectations: Quality refers to how well-designed, well-made, and national goods are and the degree of competence with which services are organised and delivered.

- Quality 'expectations' are the standards that customers anticipate a good or service will have.
- Customer expectations influence the operations process (haircut)

Quality Control and Quality Assurance

Cost-based competition: In highly competitive markets, cost can shape operational processes. It is the acknowledgement that increasing prices to increase profits is not always practical, so decreasing costs is the main way to increase profits.

- Fixed costs (rent, insurance) and variable costs (labour, input) are analysed and decreased.
- Reducing costs can be done through:
 - Eliminating waste
 - Bulk by inputs
 - Provide standardised products for larger markets

		- Achieve economies of scale - Produce high-volume output - Use automated production systems. Government policies: Businesses operate in an environment that is affected by government policies. These policies impact operational processes. • Includes Work Health and Safety, Public health, environment, employment relations, etc. • Government regulation limited the number of people allowed in restaurants because of COVID. - Restaurants modified their operations by focusing on takeout.
	Corporate Social Responsibility	Corporate social responsibility: Open and accountable business actions based on respect for people, community/society and the broader environment • Doing more than just complying with laws. • Pay employees a 'living wage' instead of a minimum. • Impacts operations through where a business sources its inputs, the impacts its operations have on the community and environmental impact. Legal compliance: Business follows the law Ethical responsibility: Meets legal obligations and demonstrates values other than money - CSR motivates employers in their work, hence, more engagement and productivity. - The social conscience of business. Adoption of policies of conservation, recycling and restoration. Legal regulation: The range of laws with which a business must comply is called 'compliance' (there are people who check that businesses are complying with the law) • Racial Discrimination Act (1975) • Anti-Discrimination Act (1977) • Superannuation Guarantee Act (1992) • Environment Protection and Biodiversity Conservation Act (1999). Environmental regulation influences some operational procedures, such as manufacturing, such as carbon emissions and chemical disposal.

Environmental sustainability: Environmental sustainability means that business operations should be shaped around practices of conserving consumer resources today without compromising access to those resources for future generations.

- There is a difference between legal obligations and environmental sustainability (business takes on an optional initiative)

CASE STUDY - Zara

First opened in 1975 (initially expanded in Europe and the US), then expanded in Latin America (1992) and Asia (2005)

- Has 2249 stores: operating in 96 markets
- \$20 billion in sales
- Grew through its development of localised production in Spain, Morocco, Portugal, and Turkey
- The founder believes that production proximity and quick logistics are two contributors to global success.

CASE STUDY- McDonald's

- McDonald's partnered with Uber Eats and Menulog, which allowed customers to order food online and have that delivered.
- McDonald's also has an app that helps customers order faster, showing decreased wait times for food for customers with big orders from 50 - 100 secs to 10-15 secs.

CASE STUDY - McDonald's location

In Tecoma, Victoria, a McDonald's location proposal was refused because the local council voted against having it in the area citing zone regulations and houses of operations limits.

CASE STUDY: Mars

Mars developed five principles to lead a company:

1. Quality
2. Responsibility
3. Mutuality

		4. Efficiency 5. Freedom. - Mars has policies concerning its use of genetically modified ingredients, the health and nutrition of food, and research. - The goal is that 100% of coca will come from certified sources - The long-term goal is to reduce the total environmental impacts - Aims to improve the lives of 1 million people in their value chain, advanced science innovation and marketing in ways that help people.						
Processes	Inputs	Conversion of inputs into outputs: <table border="1"> <thead> <tr> <th>Input</th><th>Transformation Process</th><th>Output</th></tr> </thead> <tbody> <tr> <td>Materials Labour Info</td><td>Processes (quality control, package)</td><td>Good or service</td></tr> </tbody> </table> Resources used in the transformation process include: 1. Labour (physical and mental effort) 2. Energy (electricity or fuel converted into heat, lights, movement) 3. Raw materials (basic components of manufactured goods, wood) 4. Machinery and Technology (used to process raw materials integrated with tech) Transformed resources are inputs that are converted in the operations process, including: 1. <u>Materials</u>: basic elements used in the production process, which consist of: - Raw materials: Wood, metal, plastic. Intermediate - Goods: Unfinished goods that have gone through some processing (computer microchips, glass screens, paper) - For services: materials include stationery, computers, and furniture. 2. <u>Information</u>: The knowledge gained from research, investigation, and instruction It is an input because it tells us what inputs to use, what processes to use, where to get inputs from, etc.	Input	Transformation Process	Output	Materials Labour Info	Processes (quality control, package)	Good or service
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		3. <u>Customers</u>: It is their choices, desires, and preferences that are the input. E.g. customers' preferences lean towards healthier foods. This information informs us what food to produce. The operations process should be changed accordingly. Transforming resources (human resources, facilities), Inputs that do stuff to the transformed resources. 1. <u>Human resources</u>: The people involved in transforming the inputs include: - Labourers who are directly involved in creating the product - Managers who coordinate resources such as machinery (office manager). 2. <u>Facilities</u>: The plant (factory, office) and machinery used in the operations process
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Transformation Process

Conversion of inputs into outputs:

Input	Transformation Process	Output
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Volume:

Volume flexibility is when we adjust the operations processes to make more or less in response to demand.

- Lead times are the times it takes for an order to be fulfilled from the moment it is made. E.g. time to make a cheeseburger at McDonald's = 1 minute.

SHORTER LEAD TIME = GREATER CONTROL OF VOLUME (how much you make)

- If there is a sudden increase in orders, a shorter lead time ensures an effective response to the change in orders.

CASE STUDY - Maccas

Due to seasonal changes in customer demand (more ice cream in summer, more apple pies in winter), McDonald's ensures they have a short lead time and good inventory management to change the volume of products as customer demand changes.

Variety:

A mix of products made, sometimes called 'variety of choice' or 'product range'.

- The greater the variety, the greater the amount of operations processes that need to be considered. E.g. food manufacturer (each food needs a different process).

CASE STUDY - Whirlpool

Whirlpool makes a range of household appliances.

The electronic components within the range of devices are very similar, they heat and cool, and have rotating parts and timers. So, despite the apparent diversity in these products, many of them are electronically very similar.

Gantt Chart



The company behind the brands can sell a variety of products made with largely similar production processes.

Variation in Demand

A variation in demand will have a significant impact on the transformation process because an increase in demand will require more inputs, energy, human resources, use of facilities, etc.

- Businesses often try to anticipate demand to ensure that the operations process can keep up with changes in demand.

Influence of Visibility (customer contact)

Customer contact or feedback can directly affect transformation processes: the idea that customer preferences affect production decisions.

- For example, customers prefer ethical labour practices, so a business may change its transformation process from using foreign labour to local Australian labour with stronger legal protection.

Sequencing and scheduling

Sequencing = order of operations process

Scheduling = the length of time of the activities within the operations processes.

- When planning operations processes that involve activities central to transformation, scheduling tools must be used.

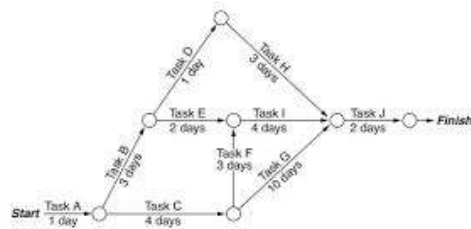
Some of these scheduling tools are:

Gantt Charts:

A Gantt Chart outlines the activities that need to be performed, the order in which they should occur, and how long each activity is expected to take.

Advantages of Gantt Charts:

- It forces a manager to plan the steps needed to complete a task and to specify the time required for each task.
- They make it easy to monitor actual progress against planned activities.



What is the minimum number of days needed to complete the project?

Critical Path Analysis:

A scheduling method or technique that shows what tasks need to be done, how long each task takes, and the order necessary to complete those tasks.

- A critical path is the shortest length of time it takes to complete all tasks necessary.
- However, the critical path is found by taking the longest path through the process.
- The reason the longest is best is because it maximises your ability to do multiple tasks at the same time, with the shorter paths, there isn't enough time to do some of the longer tasks.

The main difference between a Gantt chart and a critical path analysis is that a Gantt chart offers a timeline view of a project, whereas a critical path maps the sequence of scheduled tasks that determine a project's duration.

Technology:

Involves the use of machinery and systems that enable businesses to undertake transformation processes more effectively.

- E.g. office tech = computer, data storage, modem, phone, scanner.

CASE STUDY: The Virtual Office

Utilising video conferencing, cloud-based solutions, database social media, etc., allows businesses to conduct operations from any location at any time.

Employees can access these facilities anywhere and complete the process more effectively.

Manufacturing tech:

1. **Robotics:** highly specialised tech capable of complex tasks. Allows for precision and consistency
2. **CAD:** A Tool that allows businesses to create product possibilities in the form of 3D diagrams.
3. **CAM:** Software used to alter manufacturing process: analyses the process and makes suggestions for inputs, methods, efficiency, etc.

CASE STUDY: Manufacturing Tech

In 2017, Nike Collaborated with the world's fastest marathoner, Kipchoge to develop the Nike Zoom Vaporfly, Eliud wore the 3D-printed shoe to run in the Berlin Marathon in 2017. The shoes were super lightweight: more than could be achieved with regular manufacturing.

		Task design Involves classifying job activities in ways that make it easier for an employee to successfully perform and complete the task. Usually involves breaking down the work into a series of jobs. - This is also how a business determines what people are needed in the transformation process. - E.g. in a restaurant you would need a chef, kitchen hand, waiter, etc. to complete the transformation process. Process layout Organising the physical layout of a workplace. This depends on the manufacturing operations or services performed by the business. There are 3 main options: 1. Process layout 2. Product layout - Characterised by manufacturing a high-volume, low-variety good. Commonly, this is an assembly line. 3. Fixed position layout. - Layout for large-scale, bulky activities such as construction of buildings. The product (a building) is in one place and all the transformation processes occur there and everything comes to the site. Monitoring Measuring actual performance against planned performance, & involves the measurement of all aspects of operations. - For example, McDonald's may set a goal of average customer wait times (lead times) of no more than 120 seconds and then measure the actual wait times to determine if the goal is being achieved. - Transformation processes must be monitored for effectiveness and then improved: this ensures quality and profitability, and that goals are being met. Control Taking corrective action (redesigning the facility's layout) if goals are not being achieved. - E.g If wait times at McDonald's are more than 120 seconds on average, managers might decide
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		to modify transformation processes (change staff positioning). - Controlling compares what was intended to happen with what has occurred. - Identifies if intervention is needed. Helps to assess progress, which guides to success. Improvement: The systematic reduction of inefficiencies and waste. - Improvement is sought in time, process flows, quality, cost and efficiency. (reduce bottlenecks).					
	Outputs	Conversion of inputs into outputs: <table border="1"> <thead> <tr> <th>Input</th><th>Transformation Process</th><th>Output</th></tr> </thead> <tbody> <tr> <td>Materials Labour Info</td><td>Processes (quality control, package)</td><td>Good or service</td></tr> </tbody> </table> The result of a business's efforts is the final good/service that is provided to the customer. Most businesses provide a combination of both (food/drink is good, cooking, and serving is service). Customer Service How well a business meets and exceeds the expectations of customers in all aspects of its operations. - If a customer is dissatisfied with wait/lead times, quality or services then operations processes need a review (this info is now customer preference). Warranties A good way to assess the effectiveness of operations processes is to measure the number of warranty claims. - Warranty claims mean that a good or service has defects arising from transformation, and the operation processes need to be looked at. Provides proof.	Input	Transformation Process	Output	Materials Labour Info	Processes (quality control, package)
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Strategies	Performance Objectives	Performance Objectives: Goals that relate to particular aspects of the transformation process. The objectives help inform the choice of strategies discussed. Quality: 1. <u>Quality of design</u>: how well a good is made or a service is delivered. Uses high-quality materials, aesthetically pleasing and long-lasting. 2. <u>Quality of conformance</u>: how consistently products achieve compliance with design specifications (these specifications can be high or low, but consistent). 3. <u>Quality of service</u>: service is reliable, meets customers' needs and responds to their expectations. CASE STUDY - Apple ID The iPhone is well-known for its quality, made of high-grade materials, designed by experts and aesthetically pleasing. Reflected by the goals of Apple and the reputation they built. Their operations processes have been successful in achieving the objective of quality. Speed: Refers to the time it takes for the production process to respond to changes in demand, Speed means that a business can change the amount of input and respond to customer expectations quickly. - Goals: shorter wait times and faster processing. CASE STUDY - McDonald's Many of McDonald's critical points of operations are focused on achieving. They plan and specify times for food to be prepped, how many seconds it takes to assemble a burger, etc. Resulted in a 253 sec average on delivering food (due to tech called Dynamic - adapts digital menu) Dependability: Refers to how consistent and reliable a business's products are. Can also mean how long a good will last. Measured by predictability. CASE STUDY - The Unkillable Toyota Hilux.
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		Flexibility: Refers to how quickly the operations processes can adapt to changes in the market. Changes in market demand (how much people want) can cause pressure on capacity (how much the business can make). - Relationship between speed and flexibility. If the business can deliver goods/services quickly, it is flexible. Customisation: Most foods are standardised (to make the operation process easier, cheaper and quicker) - Sometimes, customisation, which is creating individualised products to meet specific needs, can be beneficial. - Mass customisation is a model that has the best of both worlds. A business mass-produces a product but adds customisation features to meet customer expressions. CASE STUDY - Volkswagen Volkswagen sells around 20,000 models in Australia per year - the car is mass-produced to meet this demand. However, customers are given the ability to customise colours, interiors, engines, and other features. Cost: Cost as a performative objective refers to the minimisation of expenses such that operational processes are conducted with cost reduction. Usually done through: 1. Using new equipment/better technology 2. Using inputs more efficiently 3. Reducing wastage 4. Getting discounts from suppliers 5. Managing inventory more effectively
	New Product or Service Design and Development	The operations processes of business require the creation, design and development of new products. - Enables a business to grow and attain a competitive advantage. Two different approaches determine product design and development: 1. Consumer preferences

		2. Changes and innovations in technology. Important factors in new product design and development include quality, supply chain management, output capacity and cost. A service can be: - Explicit: The application of time, expertise, skill and effort - Implicit: The feeling of being looked after. E.g. a service can be both the menu and script followed by wait-staff.					
	Supply Chain Development	Supply chain management involves integrating and managing the flow of supplies through the inputs, transformation processes, and outputs to best meet the needs of customers. Can be divided into: <table><tr><th>Sourcing (Supply Side)</th><th>Distribution and Logistics</th></tr><tr><td>Deciding where to buy materials, how much, and from which business.</td><td>Getting inputs from the source to the place where the transformation takes place, and then to the customer.</td></tr><tr><td>Sourcing other factors of production, line labour and capital.</td><td>Storing and managing the flow of goods.</td></tr></table> Logistics: Broadly refers to the distribution, but also includes transportation, storage (warehouse) and handling/packing materials. Includes: 1. <u>Distribution</u>: how we deliver goods/services to the customer (factory to warehouse, trucks) 2. <u>Storage/warehousing</u>: Finding a place to store materials/goods until they are needed. There are many costs associated with this, like rent, insurance, moving stock, and damage. 3. <u>Materials handling</u>: making sure inputs/goods are treated correctly during distribution and storage, e.g. refrigeration, and packing for fragile items.	Sourcing (Supply Side)	Distribution and Logistics	Deciding where to buy materials, how much, and from which business.	Getting inputs from the source to the place where the transformation takes place, and then to the customer.	Sourcing other factors of production, line labour and capital.
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		E-Commerce: Involves buying and selling goods and services via the Internet. About supply chain management, e-commerce is used for sourcing inputs. Includes: Business-to-Business (B2B) - A furniture-producing business will take their orders from their wood suppliers online. This may be automated so that when the stock falls below a certain level, orders are automatically set out. Ensures there are always enough inputs for operations. Business-to-customer (B2C) - Selling goods online like many online retailers (Shien, etc.) Global Sourcing: Sourcing refers to the purchasing of inputs for the transformation process. Sources or inputs are drawn from a range of suppliers. Factors that impact sourcing are: - Customer demand - Quality of inputs required - Flexibility of the supplier - Costs of one supplier vs another Global sourcing means sourcing inputs from around the world. It has advantages, e.g. fabric outside of Australia is cheaper, and can thus reduce overall costs. - *E-procurement: the use of online systems to manage supply, which allows suppliers direct access to the business's level of supplies.
	Outsourcing	Involves the use of external providers to perform business activities. Many activities can be outsourced, such as: 1. Operations such as manufacturing (Zara purchasing clothes in Bangladesh) 2. Human resources, such as employee leave and pay (the DET outsourced their HR to a business called 'Ed connect') 3. Admin work such as data entry (large accounting businesses outsource data entry to the Philippines) 4. Customer service: call centres in India/Philippines for many businesses in Aus, like Telstra.

		Advantages: 1. <u>Simplification</u>: reduces the number of activities in a business 2. <u>Efficiency and cost savings</u>: cheaper labour, regulatory differences 3. <u>Access to skills/resources is lacking in the business</u> 4. <u>Capacity to focus on business treasures</u> 5. <u>Strategic benefit</u>: avoid trade barriers Disadvantages: 1. <u>Cost of arranging</u>: can be complicated and costly to set up 2. <u>Communication and language</u>: different countries 3. <u>Loss of control</u>: may have lower standards 4. <u>Loss of domestic employment</u> 5. <u>Information security breaches</u> 6. <u>Management</u>: potential resistance to change, industrial conflict, customer backlash, manage change effectively, ensure legal obligations are fulfilled. 7. <u>Employees</u>: loss of jobs, contractual obligations, industrial action
	Technology	Businesses benefit from applying technology to their processes. The use of tech as a strategy can have a significant impact on cost reduction, and improvement in efficiency. Leading Edge: Technology that is the most advanced or innovative at a point in time, for example: 3D printing shoes at Nike. - Leading-edge technologies are created by processes and thinking. - When inventive inputs are created, new products can be made, which can change markets. Established: Already established and widely used, and accepted technology. - Include the use of computers and various software packages in managing business operations and functions.

Inventory Management

Inventory (stock) refers to the number of raw materials, intermediate or finished goods that a business has on hand at any time.

Advantages and disadvantages of holding stock:

Advantages	Disadvantages
- Consumer demand is more easily met - If a particular good runs out, another can be offered - Reduces lead time - Discount older stock to generate cash - It's an asset, so it makes your books look good	- Costs: storage, spoilage, insurance, theft damage - If stock is sitting there and not being sold, the resources could have been used for something else. - The old stock may go out of style, trend, and become unsellable (fidget spinners)

At the end of the accounting period, financial management needs to determine the value of stock sold throughout the year as well as the value of remaining stock. It helps to determine a business's gross profit. Are inventory valuation methods.

- LIFO and FIFO are what it costs us to buy the stuff, not what we sold it to customers for.

FIFO (First-In First-Out)	LIFO (Last-In First-Out)
Businesses would apply cost on a first-in-first-out basis, meaning the price for the first purchased stock applies first to stock sales: stock costs are understated, so profits are overstated. Stock may be overvalued at the end. - Lower COGS - Higher profits - Higher tax liability	Would apply cost on a last-in-first-out basis, assuming that the price for the last purchased stock is applied first to stock sales. - Increased COGS - Reduced profits - Lower tax liability - Used for non-perishable goods - Usually not the most accepted - Applied in businesses that experience a high volume of stock turnover, where stock rapidly rises

	The advantage of using LIFO is that the prices used to calculate the cost of sales, and therefore the gross profit, are more recent and therefore more closely reflect their economic value. However, in times of falling prices, LIFO overstated profits and understated gross profit.
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YZD Ltd bought stock in the following order;

1. April: 5,000 units at \$10 each
2. May: 5,000 units at \$12 each
3. June: 5,000 units at \$15 each

FIFO

During July, the business sold 11,000 units for \$20. What are the COGS using FIFO and Gross Profit?

If 11,00 units are sold 5,000 are assumed to have cost \$10 each (first cost incurred), 5,000 at \$12 each, and 1,000 at \$15.

COGS = $(5000 \times 10) + (5000 \times 12) + (1000 \times 15) = \125000

GROSS PROFIT = Revenue = $11000 \times 20 = 222000 - 125000 = \95000

LIFO

11,000 units sold, 5000 cost \$15, 5000 cost \$12, 1000 cost \$10.

COGS: $(5000 \times 15) + (5000 \times 12) + (1000 \times 10) = \145000

GROSS PROFIT: $220,000 - 145000 = \$75000$.

Just In Time (JIT)

An inventory management technique

- This means the business aims to have just enough stock to meet customer demand (no excess)
- Requires responsive operations functions with flexible processing and an ability to respond to changes in market demand
- Allows retailers to display a wide range of products
- Allows businesses to reduce in-store rentals as the amount of physical space required is lowered, reducing problems with ageing stock, or damaged goods that cost \$.

		- Disadvantage: businesses may not respond to sudden increases in demand for a product, as they do not have the supply to meet the order.
	Quality Management	Degree of excellence in a product or service. Quality management is making sure that this degree is achieved consistently. Control Reduces problems and defects in the product by using inspections at various points in the production process. - For example, most assembly line-style manufacturing processes will have a step where products are inspected for potential defects. Assurance Involves the use of a system to ensure that a set of standards is achieved. - For example, the business could ensure its repairs meet Australian quality assurance standards for the repair of large trucks. Ensures national standards are met. Help differentiate it from other repairers, potentially increasing its market share. - The difference between control and assurance is that assurance is proactive (you create a system to try to prevent issues and maintain quality). Control is more about finding issues when they happen. CASE STUDY - McDonalds The McDonald's Supplier Quality Management System is applied to businesses that supply food products to the fast food chain. BSI Group is a company that conducts food safety audits on McDonald's suppliers. Ensures consistently safe food that adheres to regulations. Improvement The process of continuously making the process more efficient and effective to maintain quality. - Quality control and assurance help a business realise areas it can improve its processes, and this cycle continues. <u>Total Quality Management</u> is more focused on continuous improvement than quality assurance. A focus on <u>continuous improvement</u> is an ongoing commitment to improving businesses' goods and

		services.s <u>Innovation</u>, <u>employee involvement</u>, and <u>quality</u> are closely aligned and indicate working processes.
	Overcoming Resistance to Change	Changes that impact businesses include: 1. External: Legislation, economy, society's expectations, technology 2. Internal: Staff initiative, technology, innovative goods and services. Changes can often bring resistance from people in the business. Financial costs associated with the following: 1. <u>Purchasing new equipment</u>: - New equipment, such as machines and the implementation of new technology, can be very expensive. - Usually, this high cost is justified as new equipment increases efficiency and profitability, BUT the high costs can cause resistance to change. CASE STUDY - Self-service checkouts From a business perspective, moving from 'staffed' checkouts to self-serve machines isn't cheap. There are set-up costs, software costs, ongoing costs, the cost of integrating the machines into the tech in place, etc. 2. <u>Redundancy payments</u>: - A loss of work arises from job skills that are no longer relevant to the workplace. - Businesses can't take away these jobs, they are required by law to make payments based on the length of time the employee worked for them, their existing wage, and unused leave. Have to pay unless they worked less than a year or worked for less than 15 employees. - It can be very expensive and cause resistance to change. - E.g. in NSW, redundancy payments for a professional like an accountant, with 10 years of employment will be \$50 000 - 80 000. 3. <u>Retraining</u>: - Cost that arises from the reorganisation of the business's internal hierarchy or the acquisition of new technology (more common)

		- For the self-service checkout, employees would have to be trained to help customers use the machines and be trained to use them: costly. 4. <u>Reorganising plant layout</u>: - The plant is the factory where the equipment is located. The way the plant is laid out is based on the transformation process. - Major changes mean that the layout will have to be reorganised, expensive. 5. <u>Inertia</u>: - A type of psychological resistance to change. People are stuck in their ways and due to feelings of uncertainty or fear of the unknown, they resist. - The effect of psychological uncertainty? - Changes that the business wishes to make will only be effective if the people implementing them are positive and competent with the implementation process. - If there is psychological reluctance to accept the change, there will be a lack of commitment and thus unsuccess.
	Global Factors	Global sourcing: Refers to businesses purchasing supplies or services from the global market. - As an operations strategy, it involves undertaking any business decisions that lead to cost advantages: low-cost inputs, supplies in high volume, increased product range, and access to products not yet available domestically. - Save money & increase customer satisfaction. Economies of Scale: A cost advantage that can be gained by increasing the size or scale of production. - It is a global factor because expanding outside a nation allows for more customers, production, and thus more economies of scale. - Allows greater efficiencies in production. Economies of scale increase profits in the long term. When expanding, profits may be reduced because of the extra costs; however, once the cost per unit is reduced, the profit margin (price - costs) increases between outputs and costs.

		Inputs are bought in bulk so it becomes cheaper to produce a hundred cars than it is to make one. Scanning and Learning: This means the business is scanning and learning business strategies from the global market and using them to improve its operations. - When a business sees an international competitor doing something effective, it copies it. Research and Development: Helps a business create leading-edge technology to produce innovative goods/services and solutions. - The Australian government provides an incentive for R&D through tax concessions.
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Links to McDonald's Case Study:

Strategic Role of Operations Management: Cost Leadership and Product Differentiation

- **Cost Leadership:** McDonald's uses standardised production processes across franchises to lower costs. Automation in cooking equipment reduces reliance on labour, ensuring cost leadership.
- **Product Differentiation:** McDonald's implements localised menu items such as the McSpicy in Australia or vegetarian options in India to differentiate in global markets.
- **Economies of Scale:** McDonald's bulk purchases from suppliers, achieving economies of scale that reduce per-unit input costs globally.

Influences on Operations Management

Globalisation

- Operates in over 100 countries, tailoring menus to local markets (e.g., Teriyaki Burgers in Japan).
- McDonald's global supply chain relies on centralised production for consistency, but flexibility for local tastes.
- Partnerships with Uber Eats and Menulog for global delivery solutions.

Technology

- The Digital Food Safety (DFS) system ensures accurate, real-time food safety checks.
- E-Production forecasting software predicts daily demand to reduce waste and increase efficiency.
- Touchscreen kiosks allow for faster, more accurate ordering in-store.

Quality Expectations

- Uses the Supplier Quality Management System (SQMS) to ensure ingredient quality across global franchises.
- Dynamic Yield technology adapts drive-thru menu boards to local customer preferences and real-time demand.
- Implementation of McCafe Barista Training Programs to maintain coffee quality.

Cost-Based Competition

- McDonald's competes with Hungry Jack's and KFC by offering budget items (e.g., loose change menu) while maintaining profitability through cost efficiencies.
- Automation in cooking and predictive maintenance on machinery reduces downtime and labour costs.

- Sourcing packaging materials in bulk to reduce unit costs while meeting sustainability targets.

Government Policies

- Compliance with Australia's Health Star Rating system on menu items.
 - Shifted to paper straws and sustainable packaging to meet environmental regulations.
 - Adapted to COVID-19 government restrictions by emphasising drive-thru, delivery, and online ordering.
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Processes in Operations

Technology in the Transformation Process

- Dynamic Yield AI in drive-thrus speeds up order processing by recommending popular items.
- MyMaccas App integration enables customisation and pre-ordering, reducing customer wait times.
- Automation in cooking equipment (e.g., burger grills with timers) ensures consistent output.

Monitoring, Control, and Improvement

- McDonald's tracks customer wait times with specific targets (e.g., under 120 seconds for drive-thru service).
- Data analytics from the MyMaccas App help monitor peak times and stock usage, improving efficiency.
- Continuous menu adjustments based on market feedback (e.g., adding plant-based burger trials).

Performance Objectives

1. Speed: Drive-thru average time reduced from 253 seconds to less than 120 seconds with new technology.
2. Dependability: High consistency due to standardised processes globally; partnerships with reliable suppliers.
3. Flexibility: Ability to switch between in-store, drive-thru, delivery, and app-based orders.
4. Customisation: MyMaccas App allows personalised meals and coffee orders.
5. Cost: Reduced operational costs through automated cooking and digital ordering systems.